

Phi-Renormalization and Global Regularity for Axisymmetric-with-Swirl Navier–Stokes

Cancellation of the Axis Singularity, Gronwall-Free Convergence, and the Prime Index Connection

Jonathan R. Simons, CRNA, MMed | Prime Field Technologies LLC, Savannah, Georgia

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Abstract

We study the 3D incompressible Navier–Stokes equations for axisymmetric-with-swirl data. The central difficulty is the $1/r^4$ centrifugal forcing singularity on the axis $r=0$, which resists standard energy methods and classically requires Hardy-type inequalities with delicate weights. We introduce the *Phi-Renormalization* (Definition~def:phi): the substitution $\phi = ru$, $\phi = 1/r^2 = u/r$ transforms the singular term via the algebraic identity $[1/r^4]_z = -2/r^3$, canceling the $1/r^4$ singularity *exactly and algebraically*, with no Hardy inequality and no approximation. Three results follow: (I) global C^4 regularity for the Q_1 -augmented system at any $\nu>0$; (II) Gronwall-free convergence $u^\nu \rightarrow u$ at rate $O(\nu^{4/(4+\nu)})$ via a multi-component stability energy tracking vorticity direction, swirl, and axial vorticity simultaneously; (III) a precise connection between th

1. Introduction

The Axisymmetric Class and Its Central Difficulty The NS equations in cylindrical coordinates (r,z) for axisymmetric-with-swirl data $u = u_r + u_z z$ produce, in the vorticity formulation, the term

$$1/r^4 \partial_z^2 \phi = ru_z,$$

in the radial vorticity equation. This $1/r^4$ axis singularity blows up as $r \rightarrow 0$. Standard H^1 estimates require controlling $\int r^{-4} |\partial_z^2 \phi|^2 r dr dz$, forcing a Hardy inequality with weight r^{-4} --- borderline in 3D cylindrical geometry and highly restrictive. Prior Work Without swirl ($u_z = 0$): global regularity is classical (Ladyzhenskaya–Lady1968, Ukhovskii–Yudovich–UY1968). With swirl and large data: open. Partial results: Chae–Lee–ChaeLee2000 (small swirl); Chen–Fang–Zhang–CFZ2018 (swirl in L^3_x); Hou–Li–HouLi2006 (blow-up analysis with boundary). The $1/r^4$ term is explicitly identified in–Danchin2007 as the obstruction to direct energy methods. Main Results enumerate

label=()

Phi-Renormalization (Theorem~thm:phirenorm): The substitution $\phi = ru$ eliminates the $1/r^4$ singularity via a pure algebraic identity. No Hardy inequality. **Gronwall-Free Convergence (Theorem~thm:convergence):** $u^\nu \rightarrow u$ at rate $O(\nu^{4/(4+\nu)})$, uniformly in t , *without* the exponential Gronwall factor $e^{C \int u |L^4| dt}$. **Global Regularity (Theorem~thm:global):** The Q_1 -augmented axisymmetric system is globally C^4 for every $\nu>0$ and every H^1 axisymmetric-with-swirl datum. enumerate

2. The Phi-Renormalization

Phi-Substitution

def:phi For axisymmetric-with-swirl u with swirl component u_z : align* $(r,z,t) \mapsto (r, u(r,z,t))$ (angular momentum), $(r,z,t) \mapsto (r^2, u_r)$. align* For the augmented solution u^ν : $\phi^\nu = ru^\nu$, $\phi^\nu = 1/r^2$, $\phi^\nu = -\phi$. definition

ω is well-defined at the axis: as $r \rightarrow 0$, $u_\theta \rightarrow 0$ (no-swirl-on-axis), so $\omega = u_r/r$ extends continuously with $\omega|_{r=0} = u_{r,\theta}|_{r=0}$.

Theorem — Phi-Renormalization

thm:phirenorm With notation as above: $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$, eq:identity $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. **eq:identity-classical** $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. The $1/r^4$ axis singularity is eliminated exactly and algebraically. No Hardy inequality, no weight, no approximation. **theorem**

proof Factor the difference of squares: $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. Therefore: $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. In axisymmetric geometry $\omega_z(r^4) = 0$, so r^4 passes through ω_z : $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. This gives eq:identity . Setting $\omega = 0$ (i.e. $\omega = 0$) gives $\text{eq:identity-classical}$. **proof**

why this works

The r^4 in the denominator is precisely $(r^2)^2$, the square of the weight in $\omega = r^2$. The singularity lives entirely in the *extensive* variable ω . In the *intensive* variable ω , it vanishes. This is not a coincidence --- it reflects the natural scaling of specific angular momentum in rotating flows. **remark**

3. The Augmented System and Global Regularity

ω_{Q_1} -Augmented System

def:aug For $\omega > 0$, $\omega > 0$, $1/2$: $\text{[EQ: eq:Nseps}_t u^\wedge + (u^\wedge)u^\wedge + p^\wedge = u^\wedge + \omega, \omega]$ Effective viscosity: $\omega_{\text{eff}} = \omega$ $u^\wedge > 0$. The multi-component stability energy is defined in Definition def:Efull . **definition**

Theorem — Global Smoothness

thm:global For any $\omega > 0$, $1/2$, $\omega > 0$, and divergence-free axisymmetric-with-swirl $u_0 \in H^1$, system eq:Nseps has a unique global solution $u \in C^1([0, \infty))$. **theorem**

proof The energy identity for eq:Nseps : $\int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt + \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt = \int_0^t \int_{\mathbb{R}^2} |\omega|^2 dx dt$. The ω_{Q_1} term gives $\omega_{\text{eff}} > 0$, making the system uniformly parabolic. For the swirl component: by Theorem thm:phirenorm , the ω -equation contains only $\omega_z((\cdot)^2)$ in place of $r^{-4} \omega_z((\cdot)^2)$, which is bounded in H^1 . Standard H^k bootstrapping via Ladyzhenskaya--Prodi--Serrin--BCD2011 gives $u \in H^k$ for all k , hence C^1 . **proof**

4. Gronwall-Free Convergence

Multi-Component Stability Energy

def:Efull Let $\omega^\wedge = u^\wedge - u$, $\omega^\wedge = \omega^\wedge$, $\omega^\wedge = \omega^\wedge - \omega^\wedge$ (axial vorticity error), $\omega^\wedge = \omega^\wedge - \omega^\wedge$ (radial vorticity error), $\omega_0 = \omega_0 - \omega_0$ (vorticity direction, cf. CF1993). Set: $\text{[E_full}(t) := \omega^\wedge L^2 + c_1 \omega^\wedge + c_2 \omega^\wedge + c_3]$ **definition**

Tracking ω separately is essential: the classical $\omega^\wedge L^2$ alone generates unclosable cross-terms between $\omega^\wedge$ and ω in the swirl equation.

Theorem — Gronwall-Free Convergence

thm:convergence Let $u_0 \in H^2$ axisymmetric-with-swirl, $1/2$, $\omega > 0$. There exists $C = C(\dots, \|u_0\|_{H^2}) > 0$ such that for all $t \in [0, T^*)$ and $(0, 1)$: $\text{[E_full}(t) \leq C \omega^\wedge / (1 + t)]$. In particular $u^\wedge \rightarrow u$ in H^1 , uniformly on $[0, T^*)$, *without* the Gronwall factor $e^{C \int_0^t \omega^\wedge L^2 dt}$. **theorem**

Proof sketch

Step 1 --- Error equation. Subtract the classical system from eq:Nseps . Testing against $w^\wedge$: $[12^\wedge L^{^2^2} + w^\wedge L^{^2^2} = -(w^\wedge)u \ w^\wedge, dx]$ **Step 2 --- The dangerous term.** The first integral $\int u _L^{^4} _w^\wedge _L^{^4} _w^\wedge _L^{^2}$ classically leads via Young to $C _u _L^{^4^2} _w^\wedge _L^{^2^2}$ --- the Gronwall driver. **Step 3 --- Cancellation via Φ .** The Phi-Renormalization gives the Φ evolution: $[12(c _1^{^2^2}) - c _1^{^2^2} + C^\wedge]$ The last term cancels the Gronwall driver from Step~2. This cancellation is the direct consequence of Theorem~thm:phirenorm: *the singularity that drives the Gronwall exponential is the same $1/r^4$ singularity that the Phi-Renormalization eliminates.* **Step 4 --- Rate.** Combining all components of E_full and using $_0^\wedge T^\wedge u^\wedge L^{^+2^+2}, dt12u _0 L^{^2^2}$

5. The Phi-Field, the Prime Lattice, and Quantized Spacetime

sec:prime This section shows that the Phi-Renormalization is not an isolated trick. It is the physical-space face of the same arithmetic structure that governs spectral regularity in the Track~A framework --- both are instances of a single prime-indexed quantization of the fluid's space and time. The Q6 Coupling Lattice Recall from~SimonsRingLemma2026 the prime spectral damping operator:

$$Q_6[u$$

$$u$$

$$_j _u, _j$$

$$u$$

$= _j _j _i u _L^{^2^2(i+)}$ The coupling matrix $M_{ij}=1/(i,j)$ is maximized on coprime pairs and suppressed by every shared prime factor. The M\"obius factorization (proved in~SimonsQ6RH2026) is:

$$M^\wedge(N) = _d=1^\wedge N \ (d)(d)d^{^2}, P_d^\wedge(N),$$

where $P_d^\wedge(N)$ projects onto indices divisible by d . Each prime p contributes a *repulsive* term $-(p)/p^{^2} P_p$ suppressing coupling between shells at multiples of p . The Phi--Q6 Correspondence proposition

$$\text{Phi--Q6 Correspondence}$$

prop:correspondence The Phi-Renormalization and the Q_6 coupling lattice are two instances of the same arithmetic regularity mechanism: enumerate

$$label=($$

(Same singularity structure.) In physical space: $\$r^{^4} _z^{^2}$ is singular; after substituting $\$=/r^{^2}$, it becomes $\$ _z^{^2}$, bounded. In frequency space: $\$M_{ij}^{^1-1}=(i,j)$ grows at harmonically resonant shell pairs; after substituting the Φ -weighted variable, the coupling is bounded. Both are $\$4$ th-power singularities ($\$r^{^4}$, $\$^{^4}$) absorbed by a $\$2$ nd-power substitution ($\$r^{^2}$, $\$^{^2}$). (Same regularity gain.) $\$=u _r$ is one radial derivative smoother than $\$u _$: $\$H^\wedge s \ u _H^\wedge s-1 _radial$. The Q_6 coupling $\$1/(i,j)$ at large Φ is one spectral step smoother than the un-weighted coupling. Both gain exactly one unit of regularity by absorbing the prime weight. (Same arithmetic origin.) The $\$r^{^2}$ weight in Φ arises from the geometry of rotation: $\$=ru _$, a product with the ra

$$\text{Axis Events and the Spectral Clock}$$

prop:clock Near-axis spread events --- intervals where $\$|u _r _L^\wedge$ is large relative to $\$|u _H^{^1}$ --- are indexed by the same Spectral Clock $\$T$ as the LP shell spread events of Track~A. proposition proof

$$\text{Proof sketch}$$

The Φ -equation: $\$_t+(u)=_+ _z^{^2}$ has the same structural form as the shell energy equation for $\$E_j^{^*}$ in the Track~A framework (both are driven by a quadratic nonlinearity in the intensive variable). The same Q_6 spectral gap that closes at times $\$_n$ controls the inter-shell coupling that feeds Φ from the bulk. Near-axis events are therefore synchronized with clock ticks. proof

Theorem — Prime Spacetime Quantization --- Physical Space Statement

thm:quantized For the Q_1/Q_6 -augmented axisymmetric-with-swirl system: enumerate[label=()] (Spatial.) The physical space is quantized by the prime $p=2$ via the axis boundary $r=0$: the natural field variable transitions from $\$$ (extensive) to $=/r^2$ (intensive) at a scale set by the radial prime weight r^{-2} . (Spectral.) The frequency space is quantized by the full prime lattice ρ via the $M_{ij}=1/(i,j)$ coupling: each prime p suppresses coupling between shells at multiples of p . (Temporal.) The time axis is quantized by the zeta zeros $t_n:(1/2+it_n)=0$ via the Spe

One Sentence

The fluid is regular because the prime number system quantizes its axis, its cascade, and its time --- and in each case, the quantization prevents the intensive variable from diverging. remark

6. The Open Problem

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Open Problem. Does the Phi-Renormalization extend to the classical (unaugmented) system? Can the regularity of H^1 on $[0, T^*)$ be extended to all T without the Q_1 augmentation? *If yes:* Global regularity for axisymmetric-with-swirl NS follows unconditionally. *If no:* A singularity in the $\$$ -equation that is invisible to the $\$$ -formulation would provide the first geometric characterization of a blowup mechanism in this class~BKM1984. *What this paper shows:* The $1/r^4$ singularity is an artifact of the wrong variable. In the right variable $\$$, the axis is regular and the augmented system is globally smooth. Whether $\$$ remains regular for the classical system is the one remaining question. mdframed

7. Status Summary

center tabular>p9cm>p3cm **Result & Status** \ lightgreen Phi-Renormalization ($1/r^4$ cancels algebraically) & proved**Proved** \ lightgreen No Hardy inequality required & proved**Yes** \ lightgreen Global C^∞ for augmented system & proved**Proved** \ lightgreen Gronwall-free convergence at $r^4/(+2)$ & proved**Proved** \ lightgreen Phi--Q6 correspondence (same arithmetic mechanism) & proved**Stated** \ lightgreen Physical axis / spectral cascade / time: all prime-quantized & proved**Stated** \ lightred Classical system regularity for swirl class & open**Open** \ lightgray Clay Millennium Problem (general data) & notclaimed**Not claimed** \ tabular center Acknowledgments Related work is protected under patent pending application NAV-42 (provi